

Class XI Workbook Solution

Question 1

```
import java.util.Scanner;

class Library
{
    String name, author;
    double price, d, r, total_amt;

    Library()
    {
        name = "";
        d = 0.0;
        r = 0.0;
        total_amt = 0.0;
    }

    void accept()
    {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter book, author's name, cost, no. of days taken to
return");
        name = sc.nextLine();
        author = sc.nextLine();
        price = sc.nextInt();
        d = sc.nextInt();

    }

    void compute()
    {
```

```

double noed = d-5;
if(noed <= 5)
    r = 2 * noed;
else if(noed <= 10)
    r = 3 * noed;
else
    r = 5 * noed;
total_amt = (2/100.0 * price * d) + r;
}
void show()
{
    System.out.println("Book Name: " + name);
    System.out.println("Author Name: " + author);
    System.out.println("Book Price: " + price);
    System.out.println("Total No. Return Days : " + d);
    System.out.println("Fine Amount: " + r);
    System.out.println("Total Amount: " + total_amt);
}
static void main()
{
    Library L = new Library();
    L.accept();
    L.compute();
    L.show();
}
}

```

Question 2

```

class employee

```

```

{

```

```
String empname;
```

```
int empcode;
```

```
double basicpay;
```

```
employee()
```

```
{
```

```
    empname = "";
```

```
    empcode = 0;
```

```
    basicpay = 0;
```

```
}
```

```
employee(String n, int p, double q)
```

```
{
```

```
    empname = n;
```

```
    empcode = p;
```

```
    basicpay = q;
```

```
}
```

```
double salarycal()
```

```
{
```

```
    double HRA = 30.0/100 * basicpay;
```

```
    double DA = 40.0/100 * basicpay;
```

```
    double salary = basicpay + HRA + DA;
```

```
    double SA = 0.0;
```

```
    if(empcode <= 15 && salary <= 15000)
```

```
        SA = 20.0/100 * salary;
```

```
    if(empcode > 15)
```

```
        SA = 1000;
```

```
    if(SA > 2500)
```

```
SA = 2500;
```

```
salary += SA;
```

```
return salary;
```

```
}
```

```
static void main()
```

```
{
```

```
    employee e1 = new employee();
```

```
    employee e2 = new employee("abcd", 14, 14000);
```

```
    System.out.println("Salary: " + e1.salarycal());
```

```
    System.out.println("Salary: " + e2.salarycal());
```

```
}
```

```
}
```

Question 3

```
class Emirp
```

```
{
```

```
    int n, rev;
```

```
    Emirp(int nn)
```

```
    {
```

```
        n = nn;
```

```
        rev = 0;
```

```
    }
```

```
    int isPrime(int x)
```

```
    {
```

```
        int count = 0;
```

```
        for(int i=1; i<=x; i++)
```

```
        {
```

```
            if(x % i == 0)
```

```

        count++;
    }
    if(count == 2)
        return 1;
    else
        return -1;
}

void isEmirp()
{
    int t = n;
    while(t>0)
    {
        int d = t%10;
        rev = rev*10 + d;
        t = t/10;
    }
    if(isPrime(n) == 1 && isPrime(rev) == 1)
        System.out.println(n + " is an emirp no.");
    else
        System.out.println(n + " is not an emirp no.");
}

void main()
{
    Emirp obj = new Emirp(13);
    obj.isEmirp();
}
}

```

Question 4

class TheString

```

{
    String str;
    int len, wordcount, cons;

    TheString()
    {
        str="";
        len=0;
        wordcount=0;
        cons=0;
    }
    TheString(String ds)
    {
        str = ds;
    }
    void countFreq()
    {
        len = str.length();
        str = str.toUpperCase();
        for(int i=0; i<len; i++)
        {
            char ch = str.charAt(i);
            if(ch == ' ')
                wordcount++;
            //if not space and not a vowel cons++
            else if(ch != 'A' && ch != 'E' && ch != 'I' && ch != 'O' && ch != 'U')
                cons++;
        }
        if(len > 0)

```

```

        wordcount++;
    }
    void Display()
    {
        System.out.println("Original String: " + str);
        System.out.println("Word Count: " + wordcount);
        System.out.println("Consonant Count: " + cons);
    }
    static void main()
    {
        TheString obj1 = new TheString("This is a test");
        obj1.countFreq();
        obj1.Display();
    }
}

```

Question 5

```

import java.util.Scanner;
class FiboString
{
    String x, y, z;
    int n;
    FiboString()
    {
        x = "a";
        y = "b";
        z = "ba";
    }
    void accept()
    {

```

```
Scanner sc = new Scanner(System.in);  
System.out.println("Enter value of n");  
n = sc.nextInt();  
}  
void generate()  
{  
    System.out.print(x + ", " + y + ", " + z);  
    for(int i = 4; i<=n; i++)  
    {  
        x = y; //b  
        y = z; //ba  
        z = y + x;  
        System.out.print(", " + z);  
    }  
}  
static void main()  
{  
    FiboString obj = new FiboString();  
    obj.accept();  
    obj.generate();  
}  
}
```